

11 The mass of an object can be determined by its relationship to other physical quantities.

Three relationships are shown.

$$1 \quad \frac{\text{resultant force on the object}}{\text{acceleration of the object}}$$

$$2 \quad \frac{\text{linear momentum of the object}}{\text{velocity of the object}}$$

$$3 \quad \frac{\text{weight of the object}}{\text{acceleration of the object}}$$

Which relationships are always equal to the mass of the object?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 only **D** 2 and 3 only